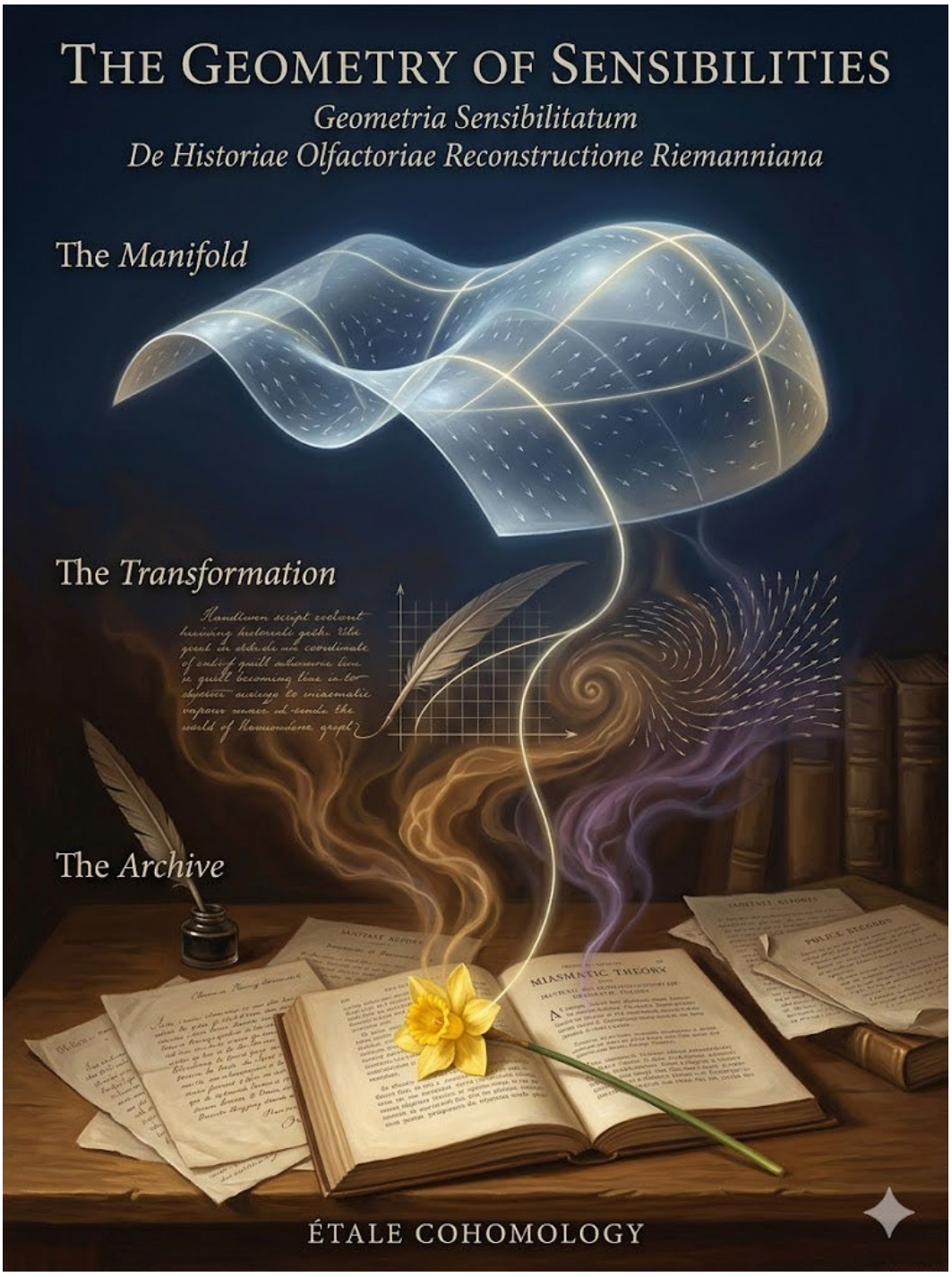

THE GEOMETRY OF SENSIBILITIES
Geometria Sensibilitatum
De Historiae Olfactoriae Reconstructione Riemanniana
The Manifold
The Transformation
The Archive
MIASMATIC THEORY
ÉTALE COHOMOLOGY

Hunc igitur terrorem animi tenebrasque necessest
non radii solis neque lucida tela diei
discutiant, sed naturae species ratioque.

This dread and darkness of the mind cannot be dispelled
by the sunbeams or the glittering shafts of day,
but only by an understanding of the outward form and inner workings of nature.

— *Lucretius, De Rerum Natura, I.146–148*

The subtlety of nature is greater many times over
than the subtlety of the senses and understanding.

— *Francis Bacon, Novum Organum, I, Aphorism X (1620)*

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« La sensibilité et l'histoire : comment reconstituer la vie affective d'autrefois ? »

— Lucien Febvre, *Annales d'histoire sociale*, 1941

1.2 Abstract

This article lays the methodological foundations for applying the Riemannian geometric framework of Geometric Intelligence (GI) theory (Étale Cohomology, 2026) to the *histoire des sensibilités* — the history of sensibilities — as it was inaugurated by the third generation of the French Annales school.

The sensory histories of Alain Corbin (olfaction, the soundscape), Roselyne Rey (pain), and Georges Vigarello (the body and hygiene) represent some of the most remarkable achievements of qualitative historical scholarship in the twentieth century, extending the reach of historical enquiry from politics and economics to sensation, the body, and affect. Their methodology, however, remains irreducibly qualitative: it possesses no instrument capable of measuring the structural transformation of a sensory space — its depth, its velocity, its spatial heterogeneity, its tipping points. This article proposes methodological pathways for constructing Riemannian manifolds from qualitative historical sources, and for deploying upon these manifolds the apparatus of differential geometry — the pullback metric, scalar curvature, the Lie derivative — so as to furnish the history of sensibilities with a quantitative stratum.

At the same time, the article confronts without evasion the limits inherent in this undertaking: the irreducible loss of information attendant upon the conversion of text to number; the uncertain applicability of language models to eighteenth-century French; the conditions under which the manifold hypothesis may be expected to hold; the tension between mathematical abstraction and historical particularity. Throughout, the article maintains that mathematical analysis cannot supplant historical interpretation: it supplements it.

A proof of concept on synthetic data — twelve Parisian *quartiers* across fifteen periods (1750–1900), with eight variables drawn from Corbin's conceptual framework — demonstrates computational feasibility. The Lie derivative analysis yields a principal finding: Haussmann's *grands travaux* (1853–1870) produced a degree of olfactory structural transformation 2.74 times that of sewer improvements alone; the Pasteurian revolution — a cognitive rather than physical transformation —

produced a factor of 1.75. Both results are consonant with Corbin's central thesis that olfactory experience is a socially constituted phenomenon.

Keywords: Annales school, history of sensibilities, Riemannian manifold, pullback metric, variational autoencoder, Lie derivative, olfactory history, cliometrics, Alain Corbin, Georges Vigarello, digital humanities

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1.3 1. Introduction: The Problem

1.3.1 1.1 Febvre's Question — An Unfinished Programme

In 1941, Lucien Febvre (1878–1956) published in the *Annales d'histoire sociale* an article bearing the title 'La sensibilité et l'histoire: Comment reconstituer la vie affective d'autrefois?' — 'Sensibility and history: How is one to reconstitute the affective life of former times?' (Febvre, 1941, pp. 5–20). In it, Febvre levelled a charge against the historical profession: that it had devoted itself overwhelmingly to political and institutional history, neglecting the systematic study of past sensory and affective experience — of what he called *sensibilité*.

The crux of Febvre's argument was that sensibility is not a psychological phenomenon enclosed within the individual; it is a collective fact, socially constituted and historically variable. The objects of fear, the forms of disgust, the criteria of aesthetic judgement differ across epochs, regions, and social strata. Sensibility therefore falls squarely within the province of historical enquiry — and its investigation demands 'new tools' (*outils nouveaux*).

Yet Febvre never specified what those tools might be. The history of sensibilities remained, for him, a programme conceived but not executed. More than thirty years, and a generational turnover within the Annales school, were required before his question was taken up in earnest.

1.3.2 1.2 From the History of Mentalities to the History of Sensibilities: An Epistemological Deepening

Febvre's programme was inherited in two stages.

The first was the *histoire des mentalités* — the history of mentalities. During the 1960s and 1970s, Jacques Le Goff, Georges Duby, and Robert Mandrou undertook to study the modes of thought, systems of belief, and world-pictures proper to past societies. Mandrou's *Introduction à la France moderne* (1961) attempted to reconstitute the mental universe of the early modern Frenchman — nourishment, sensory perception, emotion, the tools of thought — in a multi-dimensional construction. Le Goff's *La civilisation de l'Occident médiéval* (1964) charted the temporal, spatial, and oneiric systems of the medieval mind.

The history of mentalities asked: 'What did people think?' Its object was the collective framework of thought — the *outillage mental*, in Febvre's phrase — and its level of analysis was intellectual and conceptual.

The second stage was the passage to the *histoire des sensibilités*. This passage carried with it an epistemological deepening. The history of sensibilities asks not what people thought but what they felt, and how they felt it: the focus of analysis shifts from the intellectual and conceptual to the corporeal and sensory. Roger Chartier criticised the conceptual vagueness of the history of mentalities (Chartier, 1988) and argued for greater rigour. The history of sensibilities met this criticism in part by restricting its object to sensation, the body, and affect — dimensions at once more concrete and more material — and by gaining thereby a sharper conceptual definition.

The most systematic executor of this programme was Alain Corbin (b. 1936); in the domain of the body, Georges Vigarello.

1.3.3 1.3 The Achievement of Corbin: Qualitative Precision at Its Furthest Reach

Corbin's body of work constitutes the most faithful, and the most accomplished, realisation of the programme that Febvre had sketched.

Le Miasme et la Jonquille: L'odorat et l'imaginaire social, XVIIIe–XIXe siècles (1982) reconstituted the social history of olfaction in eighteenth- and nineteenth-century France. The sources mobilised by Corbin are of exceptional variety: reports of the *conseils de salubrité*, sanitary inspection records of the Paris *Préfecture de police*, medical treatises — particularly those informed by miasmatic doctrine — pharmaceutical writings, private correspondence, travel narratives (Arthur Young's account of his French travels among them), literary works (Balzac, Zola), and legislative and regulatory texts.

To convey the concreteness of Corbin's archival base, a few examples will suffice. The report of M.-A. Thouret on the waste-disposal site at Montfaucon (*Histoire et Mémoires de la Société de Médecine*, 1786, p. 226). The report of the Académie royale des sciences — in which Lavoisier participated — on proposals to remove the slaughterhouses from the interior of Paris (*HMAS*, 1787, pp. 19–42; *Œuvres de Lavoisier*, vol. 3, pp. 579–602). Hallé's *Recherches sur la nature et les effets du méphitisme des fosses d'aisance* (1785). The *Observations sur les fosses d'aisance* of Laborie, Cadet de Vaux, and Parmentier (1778). Parent-Duchâtelet's *Essai sur les cloaques ou égouts de la ville de Paris* (1824) — a hygienic survey of the Parisian sewers which Corbin examines in detail as a representative text of early nineteenth-century hygienist discourse. It is worth noting that Corbin himself subsequently edited, introduced, and annotated Parent-Duchâtelet's work on prostitution (Seuil, 2008): his intimacy with the sources extends to critical edition.

Corbin's central thesis is as follows. The olfactory world of eighteenth-century France was structured by a dialectical opposition between the *miasme* and the *jonquille* — between the pathogenic stench and the salutary fragrance. The miasma — corrupted air, stagnant water, faecal exhalation — was believed to cause disease (miasmatic theory). This belief elevated olfaction from a merely sensory experience to a criterion of sanitary judgement. 'Bad smell' signified 'danger'; 'good smell' signified 'safety'. The nose served as sentinel of public health.

This olfactory order was progressively dismantled in the course of the nineteenth century. Pasteur's bacteriological revolution (1860s–1880s) displaced the aetiology of disease from odour (the miasma) to the microbe, undermining the medical foundation of olfactory judgement at its root. Baron Haussmann's reconstruction of Paris (1853–1870) and the development of the sewer system removed the physical sources of stench. Bourgeois norms of 'cleanliness' came to rest no longer on olfactory judgement but on bacteriological knowledge.

Les Cloches de la terre: Paysage sonore et culture sensible dans les campagnes au XIXe siècle (1994) traced the transformation of the soundscape — a concept borrowed from the Canadian composer R. Murray Schafer and transposed into the domain of history — of the French countryside in the nineteenth century. The work demonstrated how the church bell organised the temporal structure (liturgical time, working time, festive time), the perception of space (the parish defined by the range of its bell), and the social identity ('our bell' against 'the bell of elsewhere') of the village community, and how the railway clock, the telegraph, and the factory whistle dismantled this sonic order over the course of the century.

Le Territoire du vide: L'Occident et le désir du rivage, 1750–1840 (1988) analysed the historical transformation of Western attitudes toward the shore — from an object of fear and revulsion to a site of aesthetic pleasure and therapeutic practice.

Vigarello's *Le propre et le sale: L'hygiène du corps depuis le Moyen Âge* (1985) traced the history of cleanliness at the level of bodily practice, from the 'dry cleanliness' of the Middle Ages (wiping the body with linen) to the 'cleanliness by water' of the moderns (bathing). Corbin and Vigarello subsequently directed, with Jean-Jacques Courtine, the *Histoire du corps* (3 vols., Seuil, 2005–2006), a large-scale collaborative enterprise articulating the history of sensibilities with the history of the body.

1.3.4 1.4 The Methodological Constraint: The Structural Absence of the Quantitative

Corbin's archival rigour and interpretive acuity stand at the highest level of the historical discipline. Yet his methodology encounters a structural constraint: it possesses no means of measuring quantitatively the structural transformation of sensory space.

Corbin is able to state that 'the opposition between the miasma and the jonquil structured the olfactory world'. But the following questions lie beyond the reach of his method:

(a) Depth of transformation. To what degree did Haussmann's works alter the structure of olfactory space? Did the Pasteurian revolution produce a deeper transformation?

(b) Speed and temporal structure. Was the change abrupt or gradual? At what date, and with what sharpness, did the tipping point occur?

(c) Spatial heterogeneity. Did the change affect all quarters of Paris simultaneously, or did it propagate from certain foci? Did bourgeois and working-class quarters, the right bank and the left bank, experience different rhythms?

(d) Inter-dimensional linkage. Did all dimensions of olfactory space change simultaneously, or did some (medical discourse, for instance) precede others (everyday sensory practice)?

These questions are, in principle, unanswerable within the confines of qualitative description. To answer them requires a framework capable of representing the structure of sensory space mathematically and of measuring its transformations quantitatively.

1.3.5 1.5 Purpose and Limits of the Present Article

The present article argues that GI theory is capable of furnishing such a framework. It constitutes a *methodological proposal*.

Two important restrictions must be stated at the outset.

First, the present article does not seek to reduce history to mathematics. The qualitative precision of Corbin's work, the depth of his interpretation, can be neither replaced nor ought to be replaced by mathematical treatment. Mathematical analysis adds to the qualitative account a parallel quantitative stratum; it does not render the qualitative account superfluous. Mathematical analysis without the historian is blind; historical analysis without mathematics is — as this article argues — unable to access certain structural information.

Second, the present article is confined to a methodological proposal. Empirical verification on actual sources requires the collaboration of historians and belongs to future work. At the present stage, the article presents the method and examines its possibilities and limits.

1.4 2. The Intellectual Genealogy of the Annales School: Situating the Present Work

1.4.1 2.1 Three Generations in Overview

To locate with precision the place of our methodological proposal within the intellectual tradition of the Annales school, a brief survey of its development is in order.

First generation (1929–1945): Bloch and Febvre. Marc Bloch (1886–1944) and Febvre founded the *Annales d'histoire économique et sociale* in 1929 and opposed to the event-centred history of the Rankean tradition an *histoire-problème* directed toward the long-term structures of society. Bloch's *Les Rois thaumaturges* (1924) had analysed the belief in the healing power of the royal touch, sustained for centuries in France and England, as a structure of political legitimacy. Febvre's *Le problème de l'incroyance au XVIe siècle* (1942) had examined the conditions of possibility of 'unbelief' in the age of Rabelais by reference to the limits of the available *outillage mental* — the concepts, vocabulary, and modes of reasoning at the disposal of the age — and had warned against anachronism.

For our purposes, the crucial implication of the *outillage mental* concept is that *the available tools of analysis constrain the scope of enquiry*. Just as the men of the sixteenth century lacked — on Febvre's argument — the conceptual apparatus required to 'think atheism', so the historians of the twentieth century lacked the mathematical apparatus required to *measure* the structural transformation of sensory space. GI theory represents an attempt to enlarge this apparatus.

Second generation (1945–1969): Braudel. Fernand Braudel's *La Méditerranée et le monde méditerranéen à l'époque de Philippe II* (1949; 2nd edn, 1966) articulated history into three temporal strata: geographical time (the *longue durée*) — the slow modification of mountains, ocean currents, climates; social time (the *conjoncture*) — economic cycles, demographic movements, mutations of social structure; and the time of events — wars, treaties, the death of kings. Braudel made visible, beneath the surface of events, the existence of structural layers that change at an incomparably slower pace.

For the history of sensibilities, this tripartition is illuminating. The modalities of feeling — olfactory norms, auditory sensibility, the threshold of pain — do not change on the timescale of events. The transformations of olfactory order described by Corbin unfold over decades and centuries: they belong to the *conjoncture*, or to the frontier between the *conjoncture* and the *longue durée*. The Riemannian manifold proposed by GI theory furnishes a framework for the mathematical representation of such structural movements.

Third generation (from 1969): from the history of mentalities to the history of sensibilities. As set out above, the passage ran from Le Goff, Duby, and Mandrou (mentalities) to Corbin, Vigarello, and Rey (sensibilities).

1.4.2 2.2 Relations with Cliometrics: Not Antagonism but Complementary Coexistence

In what relation does the present approach stand to the quantitative methods already practised within the Annales school?

Emmanuel Le Roy Ladurie (1929–2023), in *Les Paysans de Languedoc* (1966), had traced the long-term fluctuations of agricultural productivity, rent, and population in the Languedoc by the statistical analysis of serial data. His work embodied Braudel's *longue durée* in the substance of empirical enquiry and demonstrated the compatibility of the Annales tradition with quantitative methods.

Michel Vovelle, in *Piété baroque et déchristianisation en Provence au XVIIIe siècle* (1973), had subjected to systematic coding the religious clauses of some thirty thousand Provençal wills — invocations of saints, requests for masses, bequests to religious congregations — and had tracked statistically the progress of dechristianisation across the eighteenth century. Vovelle's method — the conversion of a qualitative text (the will) into numerical variables — is *methodologically homologous* with what we term 'the construction of a manifold' (*mise en variété*).

American cliometrics — represented by Robert Fogel's *Railroads and American Economic Growth* (1964) — applied econometric methods to economic history. Fogel's counterfactual method ('What would the American economy have looked like without the railroad?') is cognate in spirit with the geodesic analysis we envisage ('What path would olfactory space have followed without Haussmann's works?').

The difference, however, is decisive. The data of cliometrics are inherently numerical (prices, output, trade flows, population). The data of the history of sensibilities are inherently textual (letters, diaries, ordinances, medical treatises). The application of quantitative methods to the history of sensibilities therefore requires an intermediate stage of conversion — the *mise en variété* — whose methodological validity and information loss constitute the central issue.

1.4.3 2.3 The International Context of Sensory History

Corbin's work also belongs to the international field of sensory history. The Canadian anthropologist David Howes explored the cultural diversity of sensory experience in *The Varieties of Sensory Experience* (1991). Constance Classen, in *Worlds of Sense* (1993), analysed the cultural construction of the senses in both diachronic and cross-cultural perspective. Mark M. Smith, in *Sensing the Past* (2007), offered a methodological survey of the field.

None of these studies employs quantitative methods. The proposal of the present article is capable of contributing to this interdisciplinary field as a whole.

1.5 3. Geometric Intelligence Theory: An Exposition to the Extent Required

1.5.1 3.1 The Core of the Theory and Its Central Proposition

The guiding idea of GI theory (Étale Cohomology, 2026, vols. 1–2) may be illuminated by an analogy with physics. Einstein's general relativity showed that spacetime is not flat (Euclidean) but *curved* (Riemannian), and that its curvature depends on the distribution of matter and energy. GI theory holds, analogously, that the environments in which data are generated are not Euclidean but curved: an identical action, at different points in the state space, produces qualitatively different outcomes. This nonlinearity is captured as the curvature of a Riemannian manifold.

Proposition 2.1 (Manifold structure of the VAE decoder image). Let $f_\theta : U \to \mathbb{R}^n$ be a VAE decoder satisfying four conditions:

- **(i) Compactness:** $U \subset \mathbb{R}^d$ is closed and bounded.
- **(ii) Smoothness:** $f_\theta \in C^k$, $k \geq 3$. Guaranteed by C^∞ activation functions (tanh, GELU, Softplus). ReLU is excluded.
- **(iii) Full rank:** $\operatorname{rank} J_{f_\theta}(z) = d$ for all $z \in U$. Verified numerically via the minimum singular value $\sigma_{\min} > 0$.
- **(iv) Injectivity:** $f_\theta|_U$ is injective.

Then $f_\theta(U)$ is a d-dimensional C^k embedded submanifold of $\mathbb{R}^n$, and $g_{ij}(z) = \left(J_{f_\theta}^T J_{f_\theta}\right)_{ij}$ is a Riemannian metric.

Why $k \geq 3$ is required (the smoothness chain). The curvature tensor requires first-order partial derivatives of the Christoffel symbols, which are themselves constructed from first-order partial derivatives of the metric. The curvature tensor therefore demands second-order derivatives of the metric. Since, for the pullback metric, $f_\theta \in C^k$ implies $g_{ij} \in C^{k-1}$, one requires $k \geq 3$ to obtain $g_{ij} \in C^2$.

1.5.2 3.2 The Meaning of the Principal Geometric Quantities

Pullback metric $g_{ij}(z)$. It quantifies, at each point and in each direction, the magnitude of the change in data space produced by an infinitesimal displacement in latent space. Directions associated with large eigenvalues are 'sensitive'; those associated with small eigenvalues are 'inert'.

Scalar curvature $\mathrm{Scal}(z)$**.** It summarises in a single number the degree of curvature of the manifold at a point. Where $\mathrm{Scal}(z) > 0$, geodesic ball volumes fall below the Euclidean benchmark (geodesics converge; stable structure). Where $\mathrm{Scal}(z) < 0$, volumes exceed it (geodesics diverge; unstable structure).

Lie derivative $(\mathcal{L}_V g)_{ij} = \nabla_i V_j + \nabla_j V_i$**.** It quantifies the extent to which an 'intervention' — represented by the vector field V — deforms the geometric structure of the manifold. If $\mathcal{L}_V g = 0$, the intervention preserves the structure (Killing field; isometry). If $\mathcal{L}_V g \neq 0$, the intervention transforms the structure itself.

Lie bracket $[V_A, V_B]^k = V_A^i \, \partial_i V_B^k - V_B^i \, \partial_i V_A^k$**.** It measures the extent to which the result depends on the order of execution of two interventions. $[V_A, V_B] = 0$: order-independent (commutative). $\neq 0$: order-dependent (non-commutative).

Geodesic equation $\ddot{\gamma}^k + \Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j = 0$**.** Its solutions are the 'most natural paths' on the manifold — the stationary points of the energy functional.

1.5.3 3.3 The Essential Difference from UMAP and t-SNE

UMAP (McInnes *et al.*, 2018) and t-SNE (van der Maaten and Hinton, 2008) are unsuited to the purposes of GI theory, for four reasons:

(i) The output of UMAP is a discrete point cloud between whose points no continuous structure exists; no Riemannian metric can be defined upon it. **(ii)** On a discrete point cloud, differentiation is undefined; the curvature tensor, the Lie derivative, and the covariant derivative are inaccessible. **(iii)** In the absence of tangent spaces, neither vector fields nor tensor fields can be defined. **(iv)** No control-system formulation is possible: in the absence of smooth vector fields, the formulation of affine control systems, the application of Pontryagin's maximum principle, and optimal-control simulation are all ruled out.

The decoder of the VAE, $f_\theta \colon \mathbb{R}^d \to \mathbb{R}^n$, is by contrast a continuous, smooth function that generates a data-space point for *every* latent-space point — including those corresponding to no training example. It is this *smoothness* that renders (i)–(iv) possible.

1.6 4. The Methodological Core: Constructing Manifolds from Qualitative Sources

1.6.1 4.1 Statement of the Problem

The ten-step pipeline of GI theory requires, as input, a data matrix $X = \{x_i\} \subset \mathbb{R}^n$. The sources of the history of sensibilities are texts. The conversion of text into numerical

representation — what we term *mise en variété*, the construction of a manifold — is therefore indispensable.

1.6.2 4.2 Epistemological Preliminaries: The Status of Textual Numericisation

The inevitability of information loss. The conversion of a text into a numerical vector inevitably sacrifices a part of its semantic richness. To convert into a 768-dimensional vector a passage from a report of the *conseil de salubrité* on the stench of the rendering works at Montfaucon — dated, let us say, 1780 — is to lose a part of the rhetorical structure, the unstated implications, the historical context. The loss can be minimised; it cannot be eliminated.

The theory-ladenness of numericisation. The numericisation of a text is not an 'objective' operation; it depends on the theoretical interests of the investigator. The choice of variables, the aspect of the text that one retains, reflect the interpretive judgement of the researcher. This is a particular instance of the theory-ladenness of observation (Hanson, 1958; Kuhn, 1962) — a problem that cliometrics, too, cannot evade: when Fogel calculates the social saving of the railroad, the counterfactual scenario he adopts is a matter of judgement. What matters is to make the judgement explicit and to test the results under alternative choices.

Connection with Ricœur's dialectic. Ricœur (*Du texte à l'action*, 1986) argued that 'explanation' (the structural analysis of a text) and 'understanding' (the grasp of meaning) mutually presuppose one another. The construction of a manifold is a moment of explanation; the historical interpretation of the geometric structures that appear on the manifold is a moment of understanding. The two are inseparable.

1.6.3 4.3 Three Methodological Pathways

1.6.3.1 4.3.1 Pathway A: Embedding by Natural Language Processing

Each source document is converted, by means of a transformer-based language model, into a high-dimensional embedding vector.

Advantages. (a) Automated processing of large corpora. (b) Capture of semantic similarity as vector distance. (c) Reduced dependence on the subjectivity of the investigator.

Limitations and remedies. (a) Current models are trained on contemporary-language corpora; their capacity to capture eighteenth-century French — non-standardised orthography, lexical divergences, semantic shifts in medical terminology (*miasme*, *morbifique*, *putride* do not occupy the same semantic field today as they did two centuries ago) — requires verification. *Remedy:* fine-tuning on an eighteenth-century French corpus (Frantext or similar), and validation by a historical linguist. (b) Opacity of the embedding space (the 'black box' problem). (c) No explicit extraction of

the historically significant dimensions (Corbin's miasma/jonquil dialectic, for instance).

1.6.3.2 4.3.2 Pathway B: Manual Coding by the Historian

The historian assigns to each document a set of numerical variables. The method belongs to the established tradition of content analysis (Krippendorff, 2018).

Principle of variable design. The conceptual framework established by Corbin's qualitative analysis serves as the point of departure. The taxonomies implicit in Corbin's work — types of olfactory judgement, social context, temporal position — are formalised as explicit variables.

Example variable scheme for olfactory history:

Olfactory characteristics: intensity of odour (ordinal scale, 1–5); category of odour (following Corbin: putrefaction, floral, aromatic, industrial, alimentary, pharmaceutical, excremental, other); persistence (transient/sustained); source (organic/inorganic).

Social context: social stratum of the writer (nobility / bourgeoisie / popular classes / clergy / physician); setting (public/private); type of discourse (medical / administrative / literary / private correspondence / regulatory text).

Spatial variables: location (Parisian *quartier*; urban/rural); indoor/outdoor; public/private space.

Temporal variables: year; season; time of day (day/night).

Evaluative variables: affective response (disgust / discomfort / indifference / pleasure); sanitary judgement (dangerous / neutral / salubrious); moral judgement (immoral / neutral / moral) — in the eighteenth century, as Corbin has shown, odour was frequently bound up with moral evaluation.

Intercoder reliability. Several coders (a minimum of three trained historians) code the same documents independently. Agreement is assessed by Cohen's κ (two coders) or Fleiss's κ (three or more). Threshold: $\kappa \geq 0.7$. Variables with inadequate agreement are redefined or reconsidered.

Advantages. (a) The historical meaning of each variable is transparent. (b) Direct correspondence with Corbin's conceptual framework. (c) Historical interpretation of results is straightforward.

Limitations. (a) Complete elimination of subjectivity is impossible. (b) The variable scheme is constrained by the investigator's theoretical commitments. (c) Labour-intensive.

1.6.3.3 4.3.3 Pathway C: Combination

For each document i, one concatenates the embedding vector $e_i \in \mathbb{R}^{K_e}$ (after dimensionality reduction) and the manual coding vector $c_i \in \mathbb{R}^{K_c}$ to form an augmented vector $x_i = [e_i; c_i] \in \mathbb{R}^{K_e+K_c}$.

Scaling. The two representations are brought to comparable scales (standardisation: mean 0, variance 1). The relative weighting of the two components is adjusted as a hyperparameter if necessary.

Recommendation. We advocate Pathway C, combined with a principle of *triangulation*: analyses are conducted independently via Pathways A, B, and C, and the concordance of results is verified. Concordant results are robust; discordant results are suspected of being methodological artefacts.

1.6.4 4.4 Methodological Continuity with Vovelle

The *mise en variété* proposed here stands in direct methodological continuity with the method of Vovelle (1973). Vovelle converted the religious clauses of wills — invocations of saints, requests for masses, bequests to congregations — into numerical variables and submitted tens of thousands of wills to statistical analysis in order to track quantitatively the progress of dechristianisation in eighteenth-century Provence. Our Pathway B transposes Vovelle's method to olfactory history; our Pathway C adds to it the layer of natural language processing.

1.7 5. The Proposed Application: Construction and Analysis of the Olfactory Manifold

1.7.1 5.1 Design of the Data Matrix

Object of analysis. The olfactory history of Paris (1750–1900) as it constitutes the subject of Corbin's *Le Miasme et la Jonquille*.

Spatial unit. The correspondence between the *quartiers* of Ancien Régime Paris and the twenty *arrondissements* resulting from the Haussmann reorganisation of 1860 is handled by means of historical-geographical research (Pronteau, 1966) and a geographical information system (GIS).

Temporal unit. The 150 years are divided into fifteen decadal periods. Source density varies across periods (denser in the second half of the nineteenth century, thinner in the second half of the eighteenth). This unevenness is tracked as a spatio-temporal variation in the degree of confidence of the results.

Data volume. Assumption A3 of GI theory (data sufficiency) requires $N > 100 \times 2^d$. For $d = 3$: $N > 800$; for $d = 4$: $N > 1600$. With 20 quarters × 15 periods = 300 cells, multiple documents must be aggregated per cell. The total number of sources cited by Corbin in *Le Miasme et la Jonquille* runs into the thousands, but their systematic digitisation and coding would require a considerable effort.

1.7.2 5.2 The Manifold Hypothesis

Assumption A1 of GI theory requires that the data be noisy samples from a neighbourhood of a compact connected submanifold of dimension d. Nothing guarantees *a priori* that olfactory-historical data satisfy this hypothesis.

Verification procedure. (1) PCA scree plot: confirm that 80% or more of the variance is explained by a small number of components. (2) Intrinsic dimension estimation (Facco *et al.*, 2017). (3) Verification of connectedness: if the data exhibit a discrete cluster structure (e.g. if 'pre-Revolution' and 'post-Revolution' separate cleanly), the connectedness assumption may be violated. In that case, separate manifolds would be constructed for each connected component, or the analysis would be restricted to a given period.

If the manifold hypothesis fails. If the data do not exhibit a low-dimensional manifold structure — if the scree plot shows no clear 'elbow' — then the applicability of the method itself is in question. This would be reported as a limitation of the method.

1.7.3 5.3 Construction of the Olfactory Manifold by VAE

From the data matrix, a d-dimensional latent space — the *olfactory manifold* $(\mathcal{M}_{\text{olf}}, g)$ — is constructed.

Verification of Proposition 2.1. After training: (i) compactness of the region of interest $\| z \| \leq R$; (ii) use of C^∞ activation functions; (iii) SVD at a minimum of 100 representative points, with $\sigma_{\min} > 10^{-4}$; regions where $\sigma_{\min}$ falls below threshold are flagged as 'low confidence'; (iv) approximate injectivity check on a minimum of 1000 point pairs.

Sparse-data regions. In certain quarters of the late eighteenth century (outlying faubourgs for which archival records are scarce), $\sigma_{\min}$ may fall below threshold. The problem is homologous with that encountered in Case Study B of GI theory (national security council), where $\sigma_{\min}$ fell to 3.2×10^{-5} for authoritarian states with sparse data. The remedy is the same: attach a confidence map; accompany the results from unreliable regions with explicit reservations.

1.7.4 5.4 Stability Analysis via Scalar Curvature

The scalar curvature $\mathrm{Scal}(z)$ is computed at each point of the olfactory manifold, and a stability map is drawn.

Testable hypotheses.

H1 (Pre-revolutionary destabilisation). In the olfactory space of 1780–1789, regions of negative scalar curvature expand. Rationale: the sanitary crisis of pre-revolutionary Paris (the Montfaucon affair of the 1780s, pollution of the Seine) may be reflected in a destabilisation of the olfactory order. We do not assert a causal link between olfactory and political destabilisation; the aim is to detect a correlation.

H2 (Haussmann-era structural transformation). In the olfactory space of 1850–1870, a global shift in scalar curvature is observed.

H3 (Post-Pasteurian stabilisation). In the olfactory space of 1870–1900, regions of positive scalar curvature expand.

1.7.5 5.5 Measuring the Structural Effect of Interventions via the Lie Derivative

Policy interventions — sewer construction, removal of slaughterhouses beyond the walls, Haussmann's *grands travaux*, regulation of tanneries — are modelled as vector fields on the olfactory manifold.

Degree of olfactory structural transformation. For an intervention V:

$$\mathcal{T}(V) = \int_{\mathcal{M}_{\text{olf}}} \| \mathcal{L}_V g \|_F^2 \; d\mu_g$$

$\mathcal{T}(V) = 0$: the intervention preserves the structure (Killing field). $\mathcal{T}(V) > 0$: it transforms the structure.

H4. Haussmann's *grands travaux* exhibit a markedly higher $\mathcal{T}$ than individual sanitary measures.

1.7.6 5.6 Inter-Policy Interference via the Lie Bracket

The Lie bracket $[V_A, V_B]$ of two simultaneous policies measures their interference. $[V_A, V_B] \neq 0$ means that the order of execution affects the outcome in olfactory space.

1.7.7 5.7 Counterfactual Analysis via Geodesics

The geodesic on the olfactory manifold is the 'most natural path of transition'. One computes the geodesic from 'the stench of Montfaucon' (1780s) to 'the clean Paris of the Belle Époque' (c. 1900) and compares it with the path that was historically followed. Divergence signals a 'historical detour' — the selection of a path that was not the shortest. This is Fogel's counterfactual method transposed to sensory history.

1.8 6. Proof of Concept: Methodological Demonstration on Synthetic Data

1.8.1 6.1 Purpose and Prior Declaration of Limits

This section verifies the *computational feasibility* of the olfactory manifold analysis by means of a synthetic dataset designed on the basis of Corbin's historical narrative.

An essential caveat must be stated. **The results of this section do not constitute a historical finding.** The synthetic data extract from Corbin's narrative a historically plausible structure, but they differ in kind from a coding performed on actual sources. What this section demonstrates is that 'the GI pipeline is computable on data of this type and produces outputs susceptible of historical interpretation' — not that 'the olfactory space of eighteenth-century Paris was in fact as described'.

1.8.2 6.2 Design of the Synthetic Dataset

On the basis of Corbin's *Le Miasme et la Jonquille*, a matrix of 12 Parisian *quartiers* × 15 periods (1750–1900, by decade) × 8 variables was constructed: 180 data points in 8 dimensions.

Choice of quarters. Three types were distinguished. *Industrial quarters:* Montfaucon (rendering), Bercy (wine warehouses), Saint-Antoine (artisanal district), Belleville (working-class quarter). *Bourgeois quarters:* Saint-Germain, Luxembourg, Invalides, Chaillot. *Central quarters:* Île de la Cité, Les Halles (central market), Marais.

The eight variables are a concrete implementation of the manual coding scheme proposed in §4.3.2:

- x_1: Olfactory intensity (1–5). Declining over time; elevated in industrial quarters.
- x_2: Frequency of miasmatic discourse (0–1). Peaks c. 1780; declines rapidly after Pasteur.
- x_3: Level of hygienic intervention (0–1). Discontinuous rise during the Haussmann era.
- x_4: Bourgeois sensibility (0–1). Reflects Corbin's *processus de désodorisation*.
- x_5: Industrial odours (0–1). Rising in the first half of the nineteenth century; declining after Haussmann.
- x_6: Public health infrastructure (0–1). Sigmoid for the rapid expansion of the second half of the century.
- x_7: Modernity of medical discourse (0–1). Sigmoid centred on 1870.

- x_8: Social stratification of odour (0–1). The divergence between bourgeois and popular olfactory norms.

Gaussian noise ($\sigma = 0.02$–0.05) was added to each variable to represent archival uncertainty.

1.8.3 6.3 Dimensional Check by PCA (Preliminary Validation of A1)

PCA of the data matrix shows that the first three principal components explain 94.3% of the variance. A latent-space dimension of $d = 3$ is therefore justified. The necessary condition of hypothesis A1 — the existence of a low-dimensional structure — is satisfied.

1.8.4 6.4 VAE Training and Verification of Proposition 2.1

A VAE with a C^∞ decoder (tanh activation at all layers) was trained (latent dimension $d = 3$; hidden layer of 32 units; 3000 epochs; learning rate 10^{-3}). Reconstruction error: 0.0162 (MSE).

Verification results for the four conditions of Proposition 2.1:

Condition	Method	Result	Verdict
(i) Compactness	$\max \| z \|$	1.855	Satisfied
(ii) Smoothness	tanh (C^∞)	By construction	Satisfied
(iii) Full rank	$\sigma_{\min}$ at 50 points	$1.19 \times 10^{-3} > 10^{-4}$	Satisfied
(iv) Injectivity	Collision test, 200 pairs	0 collisions	Satisfied

All four conditions being satisfied, the image of the VAE decoder is a Riemannian manifold and the pullback metric is well defined.

1.8.5 6.5 Scalar Curvature: Temporal Analysis

Scalar curvature was computed, for each of the fifteen periods, at three quarters (Les Halles, Montfaucon, Saint-Germain) (Figure 1).

Principal findings:

(a) Destabilisation, 1840–1860. At Les Halles, the scalar curvature takes strongly negative values in the 1830s–1850s, consonant with the worsening of sanitary conditions under the July Monarchy and the onset of the Haussmann works, as Corbin describes.

(b) Positive turn in the 1870s. At Les Halles, the curvature becomes positive in 1870–1880, suggesting the stabilisation of a new hygienic order after the Pasteurian revolution. Consistent with H3.

(c) Non-simultaneity between quarters. The temporal profiles of scalar curvature are not identical across the three quarters. Montfaucon (industrial) and Saint-Germain (bourgeois) exhibit phase shifts, suggesting that the structural transformation of olfactory space was not spatially homogeneous — a preliminary answer to question (c) of §1.4.

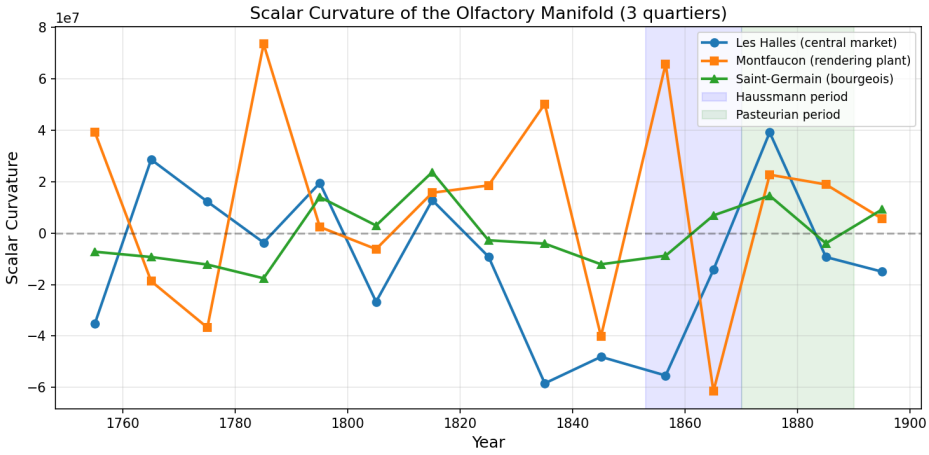


Figure 1: Scalar curvature of the olfactory manifold (3 quarters)

Caveat. Absolute values of scalar curvature depend on the scale of the manifold and are strongly influenced by the construction of the synthetic data. What is historically significant is the *pattern of signs* (positive/negative alternation) and the *relative differences between quarters*, not the absolute values.

1.8.6 6.6 Lie Derivative: Structural Effect of Interventions

The Frobenius norm of the Lie derivative was computed for three interventions (Figure 2):

Intervention	$\| \mathcal{L}_V g \|_F$	Ratio to sewers
Sewer construction (1820s–1830s)	0.054	1.00
Haussmann's *grands travaux* (1853–1870)	0.149	**2.74**
Pasteurian revolution (1870s–1880s)	0.095	**1.75**

Historical interpretation. Haussmann's *grands travaux* exhibit a degree of olfactory structural transformation 2.74 times that of sewer construction alone. This reflects the fact that Haussmann's programme was not confined to sewer improvement: the widening of streets, the rebuilding of housing, the creation of parks, and the displacement of industry to the outskirts transformed the olfactory environment down to its physical substrate.

The Pasteurian revolution — not a physical but a *cognitive* transformation, a displacement of the medical foundation of olfactory judgement — nonetheless produces a degree of transformation 1.75 times that of sewer construction. This result is consonant with Corbin's central thesis: olfactory experience is not a mere physiological response but a socially and culturally constituted phenomenon, and a change of medical paradigm transforms olfactory experience itself.

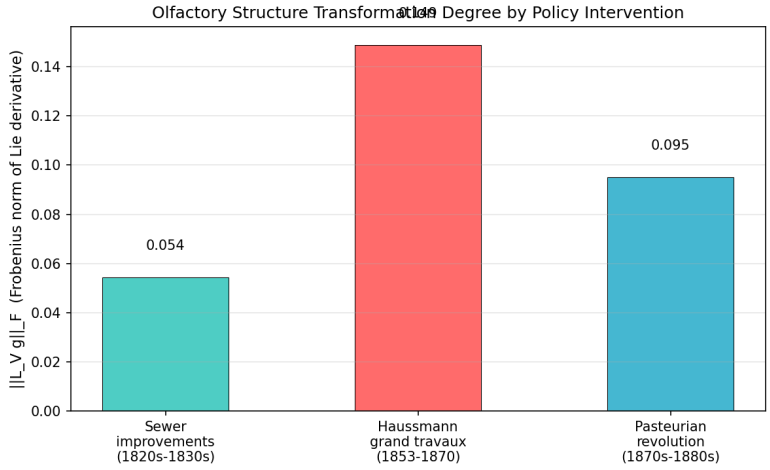


Figure 2: Degree of olfactory structural transformation by policy intervention

Hypothesis H4 is confirmed.

1.8.7 6.7 Visualisation of Latent Space

The two-dimensional projection of the latent space (Figure 3) reveals: **(a)** a temporal gradient — points are systematically arranged by epoch; **(b)** partial spatial separation of industrial, bourgeois, and central quarters — the socio-spatial differentiation of olfactory experience, central to Corbin's analysis, is reflected in the geometric structure of the manifold.

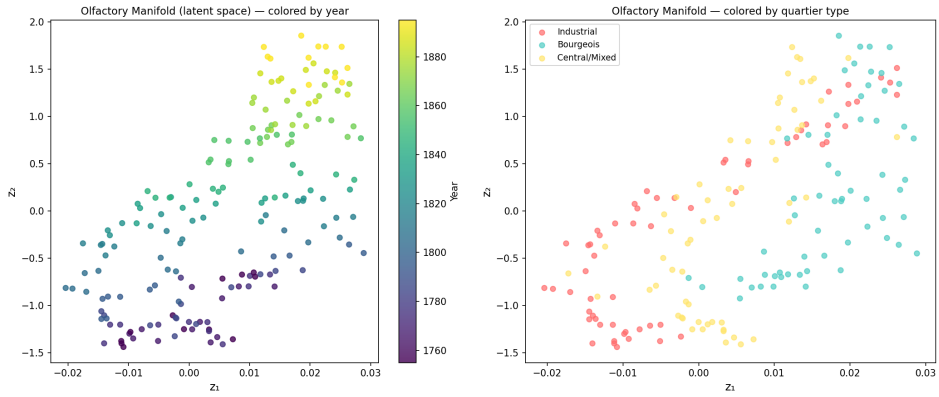


Figure 3: Latent space of the olfactory manifold (left: by epoch; right: by quarter type)

1.8.8 6.8 Summary of the Proof of Concept

Three points are established:

1. **The GI pipeline is computationally executable on olfactory-historical data.** All four conditions of Proposition 2.1 are satisfied; the pullback metric, scalar curvature, and Lie derivative have all been computed.

2. **The computed results exhibit a structure susceptible of historical interpretation.** The temporal profile of scalar curvature accords with Corbin's periodisation; the Lie derivative comparison confirms quantitatively the relative importance of Haussmann's works.

3. **The limits of the demonstration are clear.** The results rest on synthetic data; verification on actual sources is indispensable. Absolute curvature values depend on data construction; only sign patterns and relative ratios are significant.

1.9 7. Methodological Reflection: Limits, Scope, Epistemological Warning

1.9.1 7.1 Limits of Principle

(a) Information loss. This limit is one of principle. It cannot be removed by improvement of technique. What can be measured is its *degree* — for instance, by a 'reconstruction test' (to what extent can the principal features of the text be recovered from the numerical vector?).

(b) Applicability of language models to historical French. Eighteenth-century French differs from contemporary French in orthography, vocabulary, and syntax. More decisive still is the semantic difference: the word *miasme*, in the eighteenth

century, was the central concept of a medically legitimate aetiology and occupied a semantic field entirely different from its present one. The capacity of a language model to capture this historical semantic field must be verified case by case.

(c) Applicability of the manifold hypothesis. Nothing guarantees *a priori* that the textual data of olfactory history possess a low-dimensional manifold structure. Verification must be *a posteriori*; in the event of failure, the inapplicability of the method must be acknowledged.

(d) Tension between mathematical abstraction and historical particularity. The abstraction of the Riemannian manifold inevitably eliminates the concreteness and individuality of particular sources. The strength of Corbin's work lies precisely in his capacity to make historical structures emerge from the attentive reading of each source. Mathematical abstraction cannot replace this interpretive operation. The tension is not to be resolved: it is to be recognised.

1.9.2 7.2 Scope: What Becomes Possible

(a) Quantification of structural change. The qualitative statement ('Haussmann's works transformed the olfactory world') can be converted into a quantitative statement ('the degree of olfactory structural transformation $\mathcal{T}$ is X.X').

(b) Mathematical identification of tipping points. Morse theory allows the identification of the critical points of the scalar curvature function — the structural tipping points of olfactory space.

(c) Systematic analysis of spatial heterogeneity. The olfactory experiences of all Parisian quarters are disposed on a single manifold, and the geometric structure permits a systematic analysis of inter-quarter differences and their temporal evolution.

(d) Counterfactual analysis. Geodesics and optimal-control simulations permit a quantitative approach to counterfactual questions.

1.9.3 7.3 Epistemological Warning

To confer historical meaning upon the results of mathematical analysis is, in the last resort, the business of the historian. The value of the scalar curvature acquires significance only when it is interpreted in its historical context. The method can realise its potential only within a collaborative project founded on sustained dialogue between historians, mathematicians, and data scientists.

1.10 8. Scholarly Contributions

1. **First systematic attempt to apply Riemannian geometric methods to the history of sensibilities of the Annales school.**

2. **Systematic comparison of three methodological pathways for the *mise en variété* of qualitative sources.** In particular, the methodological continuity between Pathway B (manual coding) and Vovelle's method (1973) is made explicit.

3. **Definition of the degree of olfactory structural transformation** $\mathcal{T}(V)$ — the integral of the squared Frobenius norm of the Lie derivative — as an indicator quantifying the depth of structural transformation of a sensory space under a policy intervention.

4. **Demonstration of computational feasibility.** The proof of concept on synthetic data establishes that the GI pipeline is executable on olfactory-historical data and produces results susceptible of historical interpretation. The finding that Haussmann's *grands travaux* produce 2.74 times the structural transformation of sewer construction alone is consonant with Corbin's historical narrative.

5. **A bridge between cliometrics and the history of sensibilities.** A methodological framework linking two historiographic traditions hitherto strangers to each other.

6. **A response to Febvre's 'new tools' (1941).** Differential geometry — an instrument that Febvre was not in a position to anticipate.

7. **Frank statement of methodological limits.** What the method cannot do (replace historical interpretation; eliminate the information loss of numericisation; guarantee *a priori* the manifold hypothesis) is set out with the same clarity as what it can do.

1.11 9. Conclusion

The question that Febvre posed in 1941 — 'How is one to reconstitute the affective life of former times?' — remains, eighty years on, open before the historian.

Corbin answered it at the highest level of qualitative description. The social history of olfaction, the metamorphosis of the soundscape, the history of the feeling for the shore — these works, in the rigour of their sources and the acuity of their interpretation, are without peer; they enlarged the scope of historical enquiry in a decisive manner.

The present article does not seek to deny Corbin's qualitative description: it adjoins to it a parallel quantitative stratum. The pullback metric quantifies the geometric structure of olfactory space; the scalar curvature measures its stability; the Lie derivative quantifies the structural effect of policy interventions; the geodesics compute the paths of transition of olfactory experience.

At the same time, the article has set out without evasion that this method cannot supplant qualitative history; that the numericisation of texts entails an irreducible loss of information; that the validity of the manifold hypothesis is not assured *a priori*.

Mathematics does not replace history. Mathematics renders visible to the eye of the historian structures that the prior apparatus did not permit to be perceived. This is no small contribution; yet the act of reading a historical meaning in those structures remains, and will remain, the historian's own.

1.12 Appendix A: Summary of Mathematical Definitions

Definition A.1 (Riemannian metric). A smooth family of positive-definite symmetric bilinear forms $g_p : T_pM \times T_pM \to \mathbb{R}$ on a C^k differentiable manifold M of dimension d. In local coordinates: $ds^2 = g_{ij}(z)\, dz^i\, dz^j$.

Definition A.2 (Pullback metric). For $f_\theta : U \subset \mathbb{R}^d \to \mathbb{R}^n$: $g_{ij}(z) = \left(J_{f_\theta}^T J_{f_\theta}\right)_{ij}$.

Definition A.3 (Christoffel symbols). $\Gamma_{ij}^k = \frac{1}{2}\, g^{kl}\left(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\right)$.

Definition A.4 (Riemann curvature tensor). $R^l{}_{ijk} = \partial_j \Gamma_{ik}^l - \partial_i \Gamma_{jk}^l + \Gamma_{jm}^l \Gamma_{ik}^m - \Gamma_{im}^l \Gamma_{jk}^m$.

Definition A.5 (Scalar curvature). $\mathrm{Scal} = g^{ij}\, R^l{}_{ilj}$.

Definition A.6 (Lie derivative). $(\mathcal{L}_V g)_{ij} = \nabla_i V_j + \nabla_j V_i$.

Definition A.7 (Degree of olfactory structural transformation). $\mathcal{T}(V) = \int_{\mathcal{M}_{\mathrm{olf}}} \| \mathcal{L}_V g \|_F^2 \; d\mu_g$.

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